Bounded Maintenance Recoverability for Selective Lifecycle Escalation of Digital-Twin Models

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***Abstract*—A deployed digital-twin (DT) model is commonly maintained through repeated parameter-level adaptation as operating conditions evolve. This strategy implicitly assumes that the current model identity remains suitable and that local maintenance can recover acceptable service quality. That assumption can fail under persistent regime shifts or structural mismatch, where continued adaptation yields limited benefit and model-level re-instantiation becomes a relevant lifecycle option. This paper formulates *bounded fixed-identity maintenance recoverability*, which asks whether an admissible fixed-identity maintenance policy can achieve average service loss at or below a specified threshold over a given horizon and within a maintenance budget. Because future recoverability cannot be observed at decision time, a frozen estimator produces a recoverability-related admission score from causal, identity-normalized maintenance-response evidence, using supervision from finite one-shot maintenance references. A negative score admits model-level alternatives but does not force re-instantiation. Deployment-feasible alternatives then compete with retention and local maintenance under the same causal assessment data and frozen action costs. In an integrity-correction replication on 20 previously evaluated synthetic centers, the proposed policy has lower mean lifecycle service loss than frozen, unconditional-maintenance, and verification-guided local-maintenance baselines, with marginal 95% paired center-bootstrap intervals for the differences below zero. Relative to always-open search, it admits model-level search at 33.30% of stages and records 67.29% fewer candidate-search optimizer updates. The observed service-loss difference relative to always-open search is small, with a confidence interval crossing zero; this does not establish equivalence or noninferiority. Offline diagnostics still expose substantial missed-escalation flags, limiting claims about escalation timing.**

***Index Terms*—digital twin, lifecycle decision, recoverability, continual maintenance, model identity, re-instantiation, deployment constraints**

I. Introduction

Deployed digital-twin (DT) models in computing and networked infrastructures must remain useful as workloads and operating conditions evolve [1], [2], [3], [4], [5]. A common response is to adapt the parameters of the deployed model using newly available observations [6], [7], [8]. This preserves its model identity and can be effective when its learned representation remains suitable. However, a successful update at one stage does not establish that the same identity can maintain an acceptable service level over a longer horizon under limited adaptation effort. With sustained regime changes or structural mismatch, additional parameter updates may provide insufficient recovery, making a different model identity a relevant lifecycle alternative.

This distinction exposes a decision problem not resolved by drift detection or adaptation triggers alone [9], [10], [11]. Detecting a distributional change does not determine whether the current identity can meet a service requirement under a bounded maintenance policy; nor does a limited response to an individual update establish that no feasible maintenance policy remains. We therefore examine *bounded fixed-identity maintenance recoverability*: whether an admissible maintenance policy can achieve average service loss no greater than a specified threshold within a given horizon and maintenance budget, without changing model identity. Because this quantity depends on future service outcomes, it cannot be evaluated directly at online decision time. Evidence suggesting insufficient recoverability may justify *admitting* model-level alternatives, but does not by itself justify *committing* to a different identity. The latter decision still requires comparison with retention and local maintenance.

We address this lifecycle decision through causal maintenance-response evidence and selective model-level search. At each stage, a bounded probe assesses the response of the current identity to local adaptation without immediately replacing the deployed parameters. Trained on offline targets constructed from finite one-shot maintenance references, a frozen estimator produces a recoverability-related admission score from causal, identity-normalized response history. A negative score opens access to deployment-feasible model-bank alternatives; otherwise, the controller considers retention and local maintenance only. When search is admitted, eligible actions are compared on available assessment data under a frozen loss-plus-penalty rule. Only the selected action is committed, and re-instantiation resets identity-specific evidence. This separates *whether to consider* a different identity from *whether to deploy* it.

The main contributions are as follows:

- **Bounded fixed-identity maintenance recoverability.** We formulate a lifecycle-level criterion under a service threshold, maintenance horizon, budget, and admissible fixed-identity policy class. The definition distinguishes the sufficiency of continued parameter-level maintenance from the availability of model-level alternatives.
- **Maintenance-response-based admission evidence.** We construct causal, identity-normalized maintenance-response features and train an estimator using finite one-shot maintenance references. Once frozen, the estimator produces an imperfect online admission score rather than a direct evaluation of full-class recoverability.
- **Selective lifecycle escalation.** We use the score to conditionally admit deployment-feasible re-instantiation candidates while retaining local maintenance and retention as competing actions. Assessment-based action comparison determines whether an identity change is committed; identity-specific evidence is reset only after re-instantiation.

II. Related Work

A. Digital-Twin Lifecycle and Continual Maintenance

Recent DT surveys describe lifecycle synchronization, heterogeneous sensing, network-level virtualization, and model updating as central engineering challenges [1], [2], [3], [4], [5]. In edge environments, EDT-SaFL investigates semi-asynchronous federated training [6], whereas personalized federated learning supports DT construction and transfer across social IoT nodes [7]. These approaches address model learning and updates. Continual learning studies sequential knowledge retention [8]. At test time, CoTTA smooths adaptation through teacher and prediction averaging, whereas EATA filters uninformative updates and mitigates forgetting [12], [13]. RoTTA, EcoTTA, and SAR target streaming correlation, memory costs, and stability under dynamic changes, respectively [14], [15], [16]. More recent CTTA work explores representation learning, adaptive supervision, and prompt-based updating [17], [18], [19]. These studies demonstrate that maintaining changing models is an established research problem; the present work does not claim to introduce post-deployment updating or selective adaptation. Instead, it asks whether the *current model identity* can satisfy a future service threshold under bounded maintenance, and uses evidence about that question to regulate access to alternative identities.

B. Drift Detection and Adaptation Decisions

Online learning and forecasting research directly addresses nonstationarity. OneNet adjusts an ensemble under streaming concept drift [9], while RevIN and Non-stationary Transformers address distributional changes in time-series forecasting [10], [11]. PatchTST provides an alternative forecasting representation and demonstrates transferable pretraining [20]. These methods make forecasting more adaptive or accurate, but neither a change indicator nor a one-stage prediction improvement establishes the best horizon-averaged service quality achievable by bounded maintenance of a fixed deployed identity. Importantly, this distinction is conceptual rather than a claim that drift-aware methods cannot change models or adapt model ensembles. Our recoverability criterion refers to a future maintenance-policy class, while the implemented admission score is a supervised proxy obtained from a finite one-shot policy subset.

C. Model Selection, Switching, and Re-instantiation

Resource-aware model design and deployment already address architectural alternatives and operational constraints. Surveys of end–edge–cloud computing discuss heterogeneous deployment trade-offs [21]; EfficientFormer and EfficientFormerV2 construct efficient model families under latency and model-size considerations [22], [23]. MCUNetV2 jointly explores neural designs and inference schedules for limited device memory [24]. Such work establishes constrained model selection and hardware-aware design as prior capabilities. Our contribution is not a novel model bank, neural architecture search operator, or proof that model switching was previously absent. Instead, we study a separate *post-deployment admission condition*: whether the response of the currently deployed identity to bounded maintenance warrants considering source-initialized, deployment-feasible alternatives. After admission, retaining the deployed state and locally adapting it remain legitimate competitors. This formulation differs from initial few-shot model instantiation, which chooses an identity for a newly configured center, and from verification-guided fixed-identity maintenance, which decides whether to commit an adapted parameter state without introducing a different identity. Neither distinction implies that these earlier mechanisms were unavailable or ineffective.

III. Problem Formulation

A. Lifecycle State and Causal Data

At lifecycle stage t, let the deployed DT model state immediately before the decision be

$$Z_t = (m_t, \theta_t), \tag{1}$$

where m_t denotes the model identity and θ_t its parameters. This represents the deployed model, not the controller's complete history-dependent state. Update data D_t^{upd} and assessment data D_t^{assess} are visible to the online controller before the action a_t is selected. Current-stage service data D_t^{srv} are withheld from the decision interface until the action is committed and sealed. The causal order is

$$D_t^{\mathrm{upd}}, D_t^{\mathrm{assess}} \prec a_t \prec D_t^{\mathrm{srv}}. \tag{2}$$

Past service observations may subsequently enter a later stage's update data; the restriction concerns future information at the current decision.

The available lifecycle actions are

$$a_t \in \{\mathrm{retain}, \mathrm{maintain}, \mathrm{reinstantiate}\}. \tag{3}$$

Retention preserves the current model and parameters. Local maintenance commits an adapted parameter state while retaining the model identity. Re-instantiation may commit a different model identity and its adapted parameters after candidate comparison. The selected action is committed once to produce the next deployed state,

$$Z_{t+1} = \mathcal{T}_t(Z_t, a_t; D_t^{\mathrm{upd}}, D_t^{\mathrm{assess}}). \tag{4}$$

Stage-t service is then evaluated using Z_{t+1} on D_t^{srv}. The transition notation includes the applicable frozen candidate construction and action-commitment rules.

B. Bounded Fixed-Identity Maintenance Recoverability

Let $H \in \mathbb{N}_+$ be a maintenance horizon and $B_M \geq 0$ a maintenance budget. Let $\mathcal{A}_M = \{\mathrm{retain}, \mathrm{maintain}\}$ contain the fixed-identity actions. The admissible causal policy class is

$$\begin{aligned}\Pi_t^M(H, B_M) = \big\{\pi : m_{t+h}^{\pi} = m_t, \quad 1 \leq h \leq H, \\ \sum_{h=0}^{H-1} c_M(a_{t+h}^{\pi}) \leq B_M\big\},\end{aligned} \tag{5}$$

where $a_{t+h}^{\pi} \in \mathcal{A}_M$, c_M measures maintenance expenditure, and $c_M(\mathrm{retain}) = 0$. Each policy observes only the information

available by its decision time. For $\pi \in \Pi_t^M(H, B_M)$, define the horizon-averaged service loss

$$\mathcal{J}_t^M(\pi; H) = \frac{1}{H} \sum_{h=0}^{H-1} \mathcal{L}_{t+h}^{\mathrm{srv}}(m_t, \theta_{t+h+1}^{\pi}). \tag{6}$$

Given a service-loss threshold τ_{svc}, bounded fixed-identity maintenance recoverability is

$$\mathcal{R}_t^M(H, B_M) = \tau_{\mathrm{svc}} - \inf_{\pi \in \Pi_t^M(H, B_M)} \mathcal{J}_t^M(\pi; H). \tag{7}$$

A positive value implies that some admissible policy achieves average service loss below the threshold. A negative value means that no policy in the specified class reaches the threshold. At equality, a feasible policy must attain the infimum for recoverability to hold.

This definition is conditional on the policy class, horizon, budget, and threshold. A finite subset $\widehat{\Pi}_t^M \subseteq \Pi_t^M$ can provide a restricted hindsight reference, but failing to reach the threshold within that subset does not establish unrecoverability over the full class. The evaluation below does not solve the full dynamic policy-class infimum.

C. Online Decision Problem

The future-dependent theoretical quantity $\mathcal{R}_t^M$ is unavailable at decision time. The implemented controller instead uses a causal, recoverability-related admission score q_t. This score is supervised by a finite-policy reference defined in Section IV-B; it is not a numerical observation or calibrated estimate of the full-class $\mathcal{R}_t^M$. Its frozen admission rule is

$$q_t \geq 0 \;\Rightarrow\; \text{model-level search closed}, \tag{8}$$

$$q_t < 0 \;\Rightarrow\; \text{model-level search admissible}. \tag{9}$$

The decision signal is fallible: admitting model-level search does not prove that fixed-identity maintenance is impossible, and opening the gate does not force re-instantiation.

IV. Recoverability-Guided Lifecycle Method

A. Bounded Local-Maintenance Probe

At stage t, a provisional maintenance candidate is obtained by applying 50 optimizer updates to a copy of the deployed parameters using D_t^{upd}. The deployed parent remains unchanged until action commitment. Let ℓ_t^{pre} and ℓ_t^{post} be the losses of the parent and provisional candidate on the same D_t^{assess}. The relative maintenance gain is

$$g_t = \frac{\ell_t^{\mathrm{pre}} - \ell_t^{\mathrm{post}}}{\ell_t^{\mathrm{pre}} + 10^{-12}}. \tag{10}$$

This probe supplies an assessment-based response event even when the eventual action is retention. Events are stored only within the current model-identity lifetime and cleared when a different identity is committed.

B. Identity-Normalized Recoverability Evidence

For each model-identity lifetime, define the positive reference scale

$$S_m = \max\left\{\mathrm{median}(e_1, \ldots, e_{104}), 10^{-12}\right\}, \tag{11}$$

where a causal reference segment of 200 raw observations produces 104 pointwise squared errors after the 96-observation input context. At initial deployment, this reference is taken from the frozen post-warm-up segment. Following re-instantiation, it is rebuilt using the newly selected model and the last 200 raw observations of the current assessment segment. The scale is held fixed for the resulting identity lifetime.

The current response event contains g_t and the normalized post-probe residual $r_t = \ell_t^{\mathrm{post}}/S_m$. Including the current event, short and long history lengths $K_S = 2$ and $K_L = 6$ form the six features

$$\boldsymbol{x}_t = (\bar{g}_S, \bar{r}_S, \bar{g}_L, \bar{r}_L, n_S/2, n_L/6). \tag{12}$$

Here the bars denote means over the available events within each window, and n_S, n_L are their counts. Features are standardized using fitting-only statistics. A frozen MLP with architecture $6 \to 16 \to 8 \to 1$ (two ReLU hidden layers and a linear output) produces the online admission score q_t.

The score has explicit but limited supervision. The theoretical policy-class quantity is not directly used as a training target; we instead construct a finite, one-shot hindsight reference. For offline fitting labels, let $\mathcal{K} = \{0, 10, 20, 50\}$ and $H_t = \min\{4, T - t + 1\}$, where T is the last lifecycle stage. For each $k \in \mathcal{K}$, an independent candidate $\theta_{t,+}^{(k)}$ is obtained by k updates to the same pre-decision parent on D_t^{upd}. Its parameters remain fixed while assessing the following H_t service segments:

$$\bar{L}_{t,k} = \frac{1}{H_t} \sum_{h=0}^{H_t-1} \mathcal{L}_{t+h}^{\mathrm{srv}}(m_t, \theta_{t,+}^{(k)}). \tag{13}$$

The finite-policy hindsight reference and the normalized fitting target are, respectively,

$$R_t^{\mathrm{fin}}(\tau) = \tau - \min_{k \in \mathcal{K}} \bar{L}_{t,k}, \tag{14}$$

$$y_t^{\mathrm{fit}} = R_t^{\mathrm{fin}}(\tau_{\mathrm{fit}}(c, s))/S_m. \tag{15}$$

For each fitting center c and seed s, the threshold is $\tau_{\mathrm{fit}}(c, s) = 1.25 L_{\mathrm{ref}}(c, s)$, where $L_{\mathrm{ref}}(c, s)$ is the mean service MSE over the first eight stages of a C0 reference trajectory with stage-wise 50-update local maintenance. With maintenance expenditure measured in optimizer updates, these one-shot policies form a finite subset of $\Pi_t^M(H_t, 50)$. The 50-update limit bounds each tested policy, not the total computation spent enumerating alternatives, and the reference does not optimize future maintenance schedules.

These oracle labels use future service outcomes only for offline supervision, never as online inputs. The frozen estimator regresses the normalized label from causal features. Its online output is not multiplied by S_m or adjusted for a different service threshold. Final offline diagnostics use a separate common threshold τ_{final} specified in Section V, whereas online admission retains the estimator trained with $\tau_{\mathrm{fit}}(c, s)$. A positive scale preserves the sign of a given normalized target but does not

guarantee that a learned prediction has the correct sign, nor does it remove the difference between the two thresholds.

C. Selective Admission of Model-Level Alternatives

If $q_t \geq 0$, the controller compares retention with the local-maintenance candidate only. If $q_t < 0$, it additionally admits model-level candidates from a frozen bank of eleven model identities, excluding the currently deployed identity. Each RI candidate is initialized from its frozen source checkpoint. Parameter-count and operation-count limits are checked before 50-update adaptation on D_t^{upd}. All feasible candidates are assessed on the same D_t^{assess}.

Let $u = 0.0024314418915309943$ denote the frozen reference-loss unit, computed as the median of C0 reference losses across the fitting cases. The assessment-based loss-plus-penalty scores are

$$C_t^{\mathrm{ret}} = \ell_t^{\mathrm{pre}}, \tag{16}$$

$$C_t^{\mathrm{loc}} = \ell_t^{\mathrm{post}} + 0.1u, \tag{17}$$

$$C_{t,j}^{\mathrm{RI}} = \ell_{t,j}^{\mathrm{cand}} + u. \tag{18}$$

The feasible action with the smallest score is selected, with ties resolved by retention, local maintenance, and then bank order. These scores do not directly minimize the future-service infimum in Section III-B. Reported action costs aggregate only penalties for committed local and RI actions; probe, unselected candidate-search, and inference computation are not included in that total.

D. Identity Transition and Reset

An accepted RI candidate commits both its model identity and its adapted parameters to Z_{t+1}. The controller immediately clears its identity-specific maintenance-response history and related evidence. The new reference scale is computed before the next online decision, from the selected model on the last 200 raw observations of the current assessment segment, yielding 104 squared errors. This reset prevents responses obtained under different deployed identities from being mixed.

Figure 1 summarizes the method and its two admission branches. Algorithm 1 summarizes the online decision and state transition for a lifecycle stage. Finite-policy hindsight labels used for offline estimator fitting are not part of this online procedure.

Algorithm 1 Recoverability-Guided Lifecycle Decision

Require: Pre-decision model $Z_t = (m_t, \theta_t)$; same-identity evidence $\mathcal{H}_t$; scale S_m; available $D_t^{\mathrm{upd}}, D_t^{\mathrm{assess}}$; frozen estimator, model bank $\mathcal{B}$, feasibility limits, u
Ensure: Committed Z_{t+1}, updated evidence/scale, and a_t

$\theta^{\mathrm{loc}} \leftarrow \mathrm{Adapt}(\mathrm{Copy}(\theta_t), D_t^{\mathrm{upd}}, 50)$
$\ell_t^{\mathrm{pre}} \leftarrow \mathrm{Assess}(\theta_t, D_t^{\mathrm{assess}})$
$\ell_t^{\mathrm{post}} \leftarrow \mathrm{Assess}(\theta^{\mathrm{loc}}, D_t^{\mathrm{assess}})$
$g_t \leftarrow (\ell_t^{\mathrm{pre}} - \ell_t^{\mathrm{post}})/(\ell_t^{\mathrm{pre}} + 10^{-12})$
$r_t \leftarrow \ell_t^{\mathrm{post}}/S_m$; append (g_t, r_t) to $\mathcal{H}_t$
$\boldsymbol{x}_t \leftarrow \mathrm{Features}(\mathcal{H}_t)$
$q_t \leftarrow f_\psi(\mathrm{FitStandardize}(\boldsymbol{x}_t))$
$\mathcal{C} \leftarrow \{(\mathrm{retain}, m_t, \theta_t, \ell_t^{\mathrm{pre}}),$
  $(\mathrm{maintain}, m_t, \theta^{\mathrm{loc}}, \ell_t^{\mathrm{post}} + 0.1u)\}$
if $q_t < 0$ **then**
  for all $j \in \mathcal{B} \setminus \{m_t\}$ in frozen bank order **do**
    if $\mathrm{Feasible}(j)$ **then**
      $\theta_j \leftarrow \mathrm{Adapt}(\mathrm{LoadSource}(j), D_t^{\mathrm{upd}}, 50)$
      $\ell_j \leftarrow \mathrm{Assess}(\theta_j, D_t^{\mathrm{assess}})$
      Add $(\mathrm{RI}, j, \theta_j, \ell_j + u)$ to $\mathcal{C}$
    end if
  end for
end if
$(a_t, m^*, \theta^*) \leftarrow \mathrm{MinScore}(\mathcal{C})$
  Ties: retain, maintain, then frozen bank order
Commit $Z_{t+1} \leftarrow (m^*, \theta^*)$
if $a_t = \mathrm{RI}$ **then**
  Clear identity-specific response history and evidence
  $S_{m^*} \leftarrow \mathrm{ReferenceScale}(\theta^*, \mathrm{Last200}(D_t^{\mathrm{assess}}))$
end if
Seal the action; then access D_t^{srv}
Evaluate service using Z_{t+1} on D_t^{srv}

V. Experimental Setup

A. Integrity-Correction Replication and Causal Stage Protocol

The integrity-correction replication reuses 20 originally held-out, subsequently observed synthetic computing centers (IDs 33000–33019), three seeds (7311, 7312, and 7313), four frozen scenarios (C0, C1, C3, and C4), and five lifecycle strategies. Each center–seed–scenario–method combination forms a separate 24-stage trajectory, yielding 1200 trajectories and 28,800 stage records. The corrected evaluation uses one verified, ordered 20-center batch-generation realization (F1) shared across all methods; it is not a new, previously unseen held-out evaluation.

Scenario construction. Base traces use a fixed data seed and heterogeneous monitoring configurations; the common intervention starts at raw observation 6520 (stage-9 assessment). C0 retains the native target and telemetry without an added regime intervention. C1 applies a gradual target-logit shift rising from zero to 0.8 over 5120 observations, with channel-specific delayed telemetry responses. C3 abruptly permutes and recalibrates telemetry channels and reverses target-logit deviations around the center-specific base logit. C4 retains telemetry but replaces targets according to a fixed multi-lag rule with a nonlinear interaction. Thus, C3 changes *both* monitored inputs and target behavior; C4 is a changed target-generating mechanism, not merely added noise. These synthetic mechanisms are not a proof of external generalization. The five methods are compared under shared exogenous scenario inputs, starting-state references, model-bank identities, and deployment limits within each center–seed–scenario group.

The initial timeline reserves 800 raw observations for warm-up and 200 subsequent observations for the first identity-reference scale. At stage t, the assessment segment begins at raw index $b_t = 1400 + 640(t-1)$. The update, assessment, and service segments span $[b_t - 400, b_t)$, $[b_t, b_t + 240)$, and $[b_t + 240, b_t + 640)$, respectively. Update and assessment data are accessible before action selection; the service segment is exposed to the online controller only after the selected action is committed and sealed. Each segment is windowed within its own role using a 96-observation input context, yielding 304 update, 144 assessment, and 304 service prediction targets, respectively. From the second stage onward, the update segment reuses the preceding stage's service observations; these are past data at the new decision, not future information. Stage-level

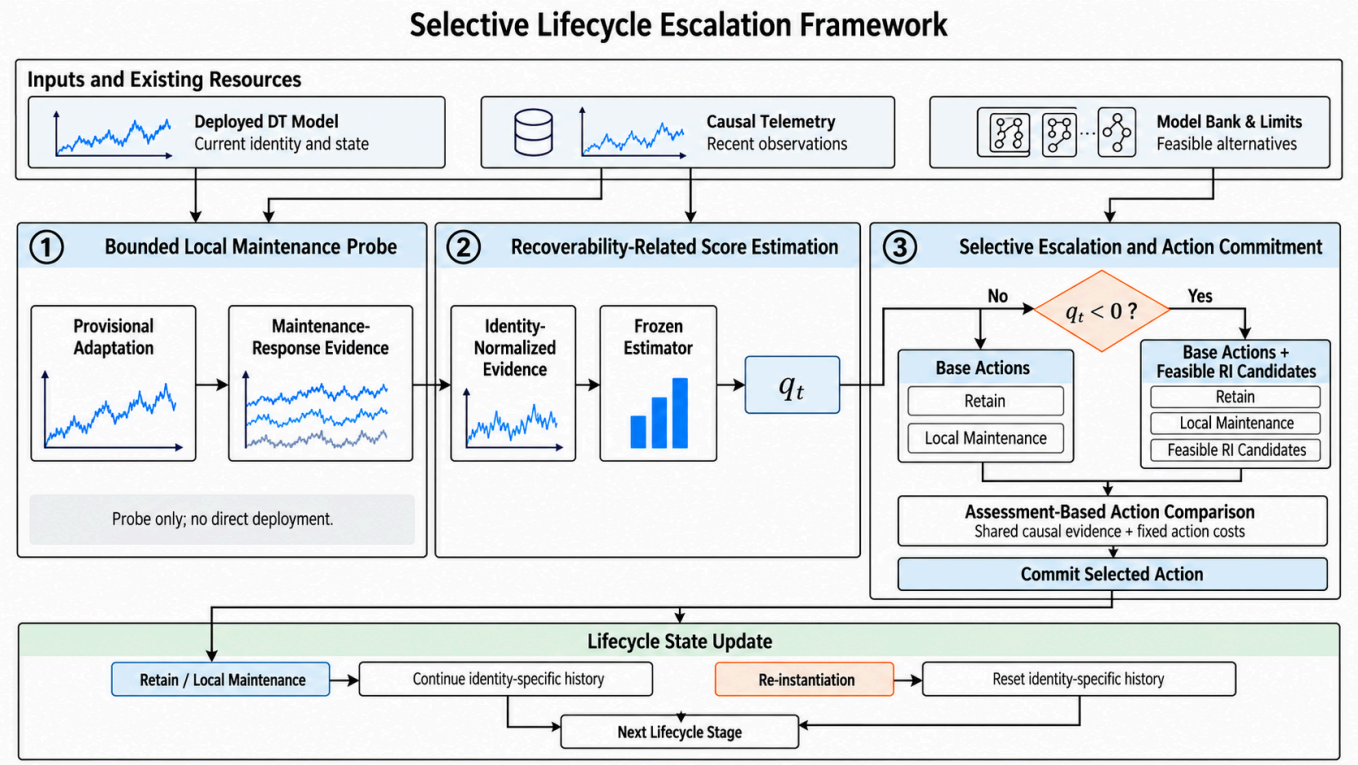


Fig. 1. Selective lifecycle escalation with score-gated RI admission and action commitment.

service loss is evaluated on the committed model state.

The estimator and feature scaler, model-bank checkpoints, candidate feasibility rules, admission boundary, action penalties, and analysis protocol remained frozen for the integrity correction. The original 1200-record inventory combined 239 full-batch and 961 singleton materializations, including one cross-method input mismatch. After that discrepancy was identified, the correction used a verified common batch20 input bundle for all methods, without pooling old trajectories into the replication. The completed replication independently certifies 1200 unique trajectories, all 240 five-method groups, 60 shared initial references, and their stage-level input and deployment-condition parity. The original results and execution evidence remain preserved. No estimator refitting, threshold retuning, model-bank change, or population reselection was performed. The controlled runtime used pandas 2.3.3, River 0.23.0, and psutil 7.2.2; full numerical equivalence to the historical runtime is not certified.

B. Compared Methods

Table I summarizes the five compared lifecycle strategies. The always-open comparator exposes model-level alternatives at every stage but does not force their commitment.

B2 verification rule. B2 forms its deployable local candidate by partitioning the 304 update-window examples into five blocks and independently adapting a copy of the current parent for 10 updates per block. It averages the five parameter deltas (the frozen $\beta = 0.1$ produces zero actually trimmed blocks) and uses six assessment error-difference blocks plus bounded historical evidence to decide whether to commit. The recent median-plus-0.5-MAD must not exceed -0.000028018231776243602, and the historical median must not exceed zero; otherwise B2 retains the parent. B2 never admits RI. The common 50-update probe is also executed but is not the B2 deployable block candidate; thus B2's online path includes 50 probe updates plus 50 block-candidate updates. Its logged local-action penalty does not enter the B2 verification acceptance condition.

C. Endpoints and Statistical Analysis

The primary endpoint is *mean lifecycle service loss*, measured as post-commit service mean squared error (MSE). Service loss is averaged over the 24 stages of each trajectory, then over seeds and scenarios within a center, and finally over the 20 centers. For each proposed-minus-baseline contrast, the observed point estimate is the difference of center-aggregated method means. Marginal percentile 95% confidence intervals use 4000 paired center-level bootstrap resamples with a shared

TABLE I
COMPARED LIFECYCLE STRATEGIES

Method	Lifecycle behavior
B0	Frozen deployed model; no maintenance or re-instantiation.
B1	Unconditional 50-update fixed-identity maintenance at every stage.
B2	Five-block local candidate with recent and historical validation; no RI.
B8	Model-level candidate search is always admissible; actions remain subject to comparison.
Proposed	Score-gated model-level search; $q_t < 0$ admits alternatives but does not force re-instantiation.

center-index matrix (seed 2904), maintaining method pairing within each resample. The resampled mean difference is not substituted for the observed point estimate. These four marginal intervals do not guarantee simultaneous family-wise coverage. Scenario-stratified analysis is reported separately. No additional unseen evaluation population was sampled after the original results became available.

Secondary measures include retain, local-maintenance, re-instantiation, and model-level-search admission rates, together with recorded feasible-candidate counts, candidate-search optimizer updates, and attributed decision-runtime measurements. The reported action-cost measure sums frozen penalties for *committed* local-maintenance and re-instantiation actions; it excludes mandatory probes, unselected candidate adaptations, and estimator inference. Search-workload counters and attributed timers are descriptive and must not be interpreted as complete end-to-end computational cost. No post-hoc weighted composite endpoint is introduced.

For offline decision diagnostics, the frozen common service threshold is $\tau_{\mathrm{final}} = 0.003039302364413743$. After each online action is sealed, a finite-policy hindsight reference $R_t^{\mathrm{fin}}(\tau_{\mathrm{final}})$ is computed from the pre-decision model identity and future service segments as specified in Section IV-B. This threshold differs from the center–seed-specific τ_{fit} used for estimator supervision and is not used to calibrate q_t or to change the operational rule $q_t < 0$. Offline labels support all-stage summaries of retain and escalation outcomes over 5760 proposed-method decisions; they are not inputs to online action selection and must not be interpreted as an online classifier-accuracy measure.

VI. Results and Analysis

A. Primary Lifecycle Service Loss

Table II reports the corrected, fully paired batch20 replication. The proposed policy achieves lower mean lifecycle service loss than B0, B1, and B2, with the respective marginal 95% paired center-bootstrap intervals entirely below zero. Its observed loss is slightly higher than B8, but the interval for that comparison crosses zero. The evidence therefore establishes neither superiority, noninferiority, nor equivalence relative to always-open search.

The observed differences use the original center-aggregated method means rather than bootstrap replicate means. The shared-resample index matrix contains 4000 draws of 20 center indices (seed 2904). Figure 2 visualizes the paired contrasts and center-level Proposed–B8 differences. In the saved center-aggregated records, 18 of 20 centers show exactly zero Proposed–B8 difference; the remaining differences are concentrated in two centers. This is a descriptive observation, not evidence of statistical equivalence or identical actions at all lifecycle stages. The B8 interval narrowly extends below zero; failure to separate methods by this interval is not an equivalence test. These are integrity-correction replication estimates from previously observed synthetic centers, not evidence from a newly unseen test population.

B. Selective Search and Action Use

Table III reports committed action frequencies and penalties; Fig. 3 contrasts model-level search with final RI commitment. B8 admits search at every stage; the proposed policy admits it at 33.30% of stages (1918 of 5760). Their observed RI counts are 105 and 104, respectively, corresponding to 1.82% and 1.81% of stages. Thus, search admission does not itself imply an identity transition, and similar overall RI counts do not establish identical switching times or identities.

Figure 3 separates model-level search admission, candidate-search optimizer updates, and committed RI actions. B8 and Proposed record 2,664,000 and 871,400 candidate-search updates, respectively (67.29% fewer for Proposed), and 105 versus 104 committed RI actions. Their recorded feasible candidate counts are 53,280 and 17,428. Attributed decision-runtime totals are 34,281 and 15,102 seconds, respectively, but include implementation and resource-scheduling effects rather than isolated end-to-end computational costs. Committed-action penalties also exclude mandatory probes, unselected candidate adaptations, and estimator inference. Reduced search does not establish that every omitted search was unnecessary.

C. Scenario-Stratified Behavior

Table IV compares Proposed and B8 under the four frozen scenarios. Their C0 mean service losses coincide in the saved results. Proposed has a slightly lower C1 observed mean, whereas B8 has lower C3 and C4 means. Scenario means alone do not establish statistical equivalence or a significant scenario-specific difference. The gate remains selective in every scenario, but this does not imply uniformly better service outcomes.

D. Offline Decision Diagnostics

The corrected offline diagnostic inventory covers 5760 proposed-method stage decisions: 407 correct-retain events, 12 unnecessary-RI events, 92 correct-escalation events, and 2100 missed-escalation flags. These named labels neither exhaust all stages nor form a complete classifier confusion matrix. Specifically, a missed-escalation flag means that RI was not committed while the finite fixed-identity hindsight recoverability margin was negative. It does not establish that a deployment-feasible RI candidate would have won assessment-based action comparison, nor that the admission gate was necessarily closed.

The substantial missed-escalation count limits claims about precise switching-time identification and full-class unrecoverability. These finite-reference labels were computed offline, after online decisions, and did not enter the online admission rule.

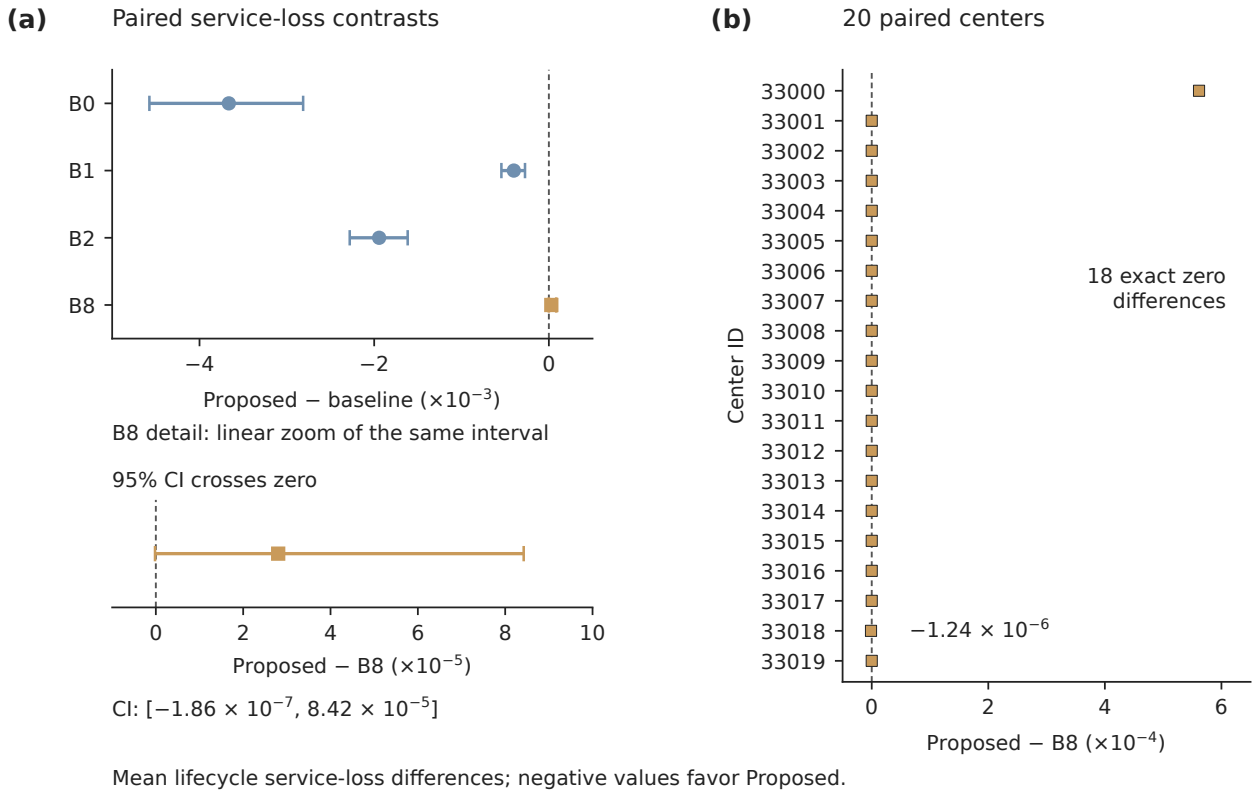


Fig. 2. Paired service-loss contrasts and center-level Proposed–B8 differences.

TABLE II
CORRECTED BATCH20 MEAN LIFECYCLE SERVICE LOSS AND PAIRED COMPARISONS

Method	Mean lifecycle service loss	Observed Proposed − baseline	Marginal 95% paired center-bootstrap CI
B0	0.006604799	−0.003664155	[−0.004573828, −0.002813402]
B1	0.003341980	−0.000401336	[−0.000543519, −0.000273248]
B2	0.004883941	−0.001943298	[−0.002280026, −0.001616486]
B8	0.002912627	+0.000028017	[-1.86×10^{-7}, $+8.42 \times 10^{-5}$]
Proposed	0.002940644	–	–

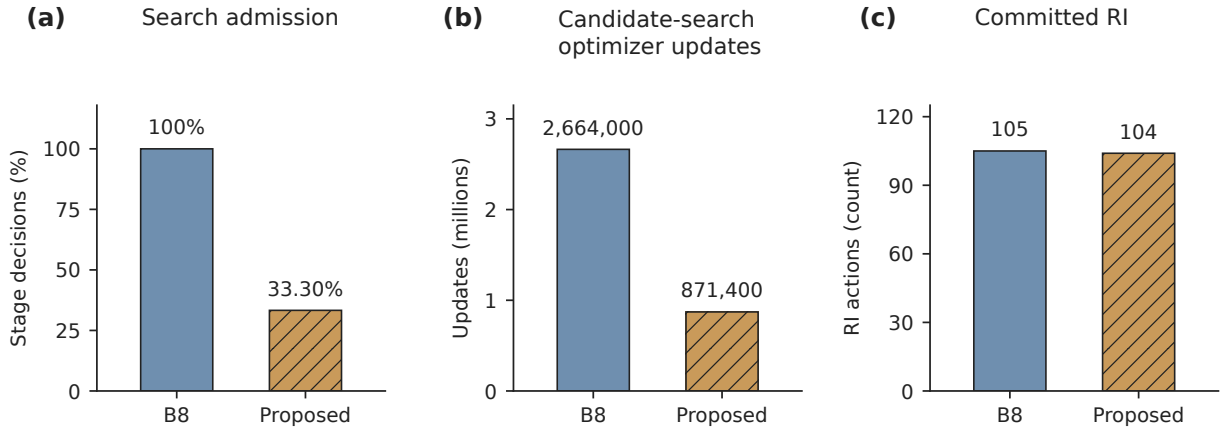


Fig. 3. Search admission, candidate-search updates, and committed RI actions.

TABLE III
COMMITTED ACTIONS, MODEL-LEVEL ADMISSION, AND ACTION PENALTIES

Method	Maintenance rate	RI rate	Action penalty	Retain rate	Gate-open rate
B0	0	0	0	1.0000	0
B1	1.0000	0	0.00583546	0	0
B2	0.2887	0	0.00168479	0.7113	0
B8	0.1240	0.01823	0.00178711	0.8578	1.0000
Proposed	0.1259	0.01806	0.00178812	0.8561	0.3330

TABLE IV
SCENARIO-STRATIFIED PROPOSED AND B8 RESULTS

Scenario	B8 loss	Proposed loss	Gate-open	RI rate
C0	0.00284816	0.00284816	0.4090	0
C1	0.00267605	0.00266895	0.2743	0
C3	0.00331095	0.00342327	0.3375	0.04375
C4	0.00281535	0.00282220	0.3111	0.02847

VII. DISCUSSION

A. Interpretation of Selective Escalation

Model-level search admission and RI commitment remain distinct. The recoverability-related score q_t admits model-level alternatives to assessment-based action comparison, but that comparison may still retain the incumbent or select fixed-identity maintenance. In the corrected replication, the gate admits search at 33.30% of stages while recording nearly the same number of RI commitments as B8. This does not establish identical switching times or identities.

The observed service-loss difference to B8 is small, and its marginal 95% interval crosses zero. This permits neither a superiority claim nor the stronger conclusion of equivalence or noninferiority. The measured candidate-search update count is substantially lower under selective admission, supporting reduced model-level search work under the recorded protocol. Attributed decision timers do not establish a full end-to-end computational advantage; action penalties cover only committed actions. The method therefore demonstrates a restricted search–service trade-off rather than uniform optimality.

B. Limits of Recoverability Evidence and Generalization

The theoretical quantity $\mathcal{R}_t^M$ is defined over a causal class of bounded fixed-identity maintenance policies. By contrast, the implemented offline reference evaluates four one-shot maintenance choices with fixed parameters over a finite future-service horizon. A negative margin over this subset cannot certify full-class unrecoverability. The frozen estimator regresses identity-normalized finite-policy targets using center–seed-specific fitting thresholds, and its score is not recalibrated to the common threshold used for offline diagnostics. The rule $q_t < 0$ is an admission heuristic, not a certified horizon-level failure of current-identity maintenance.

The correction addressed a material cross-method pairing failure in the original mixed-context run. Its uniform batch20 inputs, shared initial states and deployment conditions are certified, but the population had already been observed when this integrity-correction replication was undertaken. Numerical equivalence to the entire historical software environment remains unverified, and stochastic outcomes are not claimed to be bitwise invariant under all worker schedules. These boundaries do not invalidate the corrected paired comparison, but they preclude presenting it as a new unseen holdout or a perfect replay of every historical execution condition.

The large number of finite-reference missed-escalation flags limits claims about switching-time identification: a flag does not prove that RI would have been the best feasible action at that stage. The experiments further cover synthetic centers, four specified scenarios and a fixed model bank; they do not establish cross-domain deployment or complete computational cost. Future work should assess prospective score calibration, multi-stage maintenance references, and full-cost behavior on external traces while preserving the causal split between online decisions and hindsight diagnostics.

VIII. CONCLUSION

This paper examined when continued fixed-identity maintenance of a deployed digital-twin model may be insufficient and model-level alternatives should be considered. We defined bounded fixed-identity maintenance recoverability and developed a frozen admission estimator supervised by finite one-shot maintenance references, without claiming to solve the full theoretical policy-class optimization. The resulting score conditionally admits deployment-feasible model-level candidates to an assessment-based comparison with retention and local maintenance. In a corrected replication on 20 previously evaluated synthetic centers, the proposed policy achieved lower mean lifecycle service loss than three fixed-identity baselines and opened model-level search at 33.30% of stages, with 67.29% fewer candidate-search updates than always-open search. Relative to always-open search, the observed service-loss difference was small and its marginal 95% interval crossed zero; neither equivalence nor noninferiority was established. Substantial offline missed-escalation flags and the synthetic, post-observation evaluation scope limit claims about exact RI timing and broader generalization. Future work should evaluate prospective score calibration, multi-stage maintenance evidence and end-to-end costs on independent external traces.

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